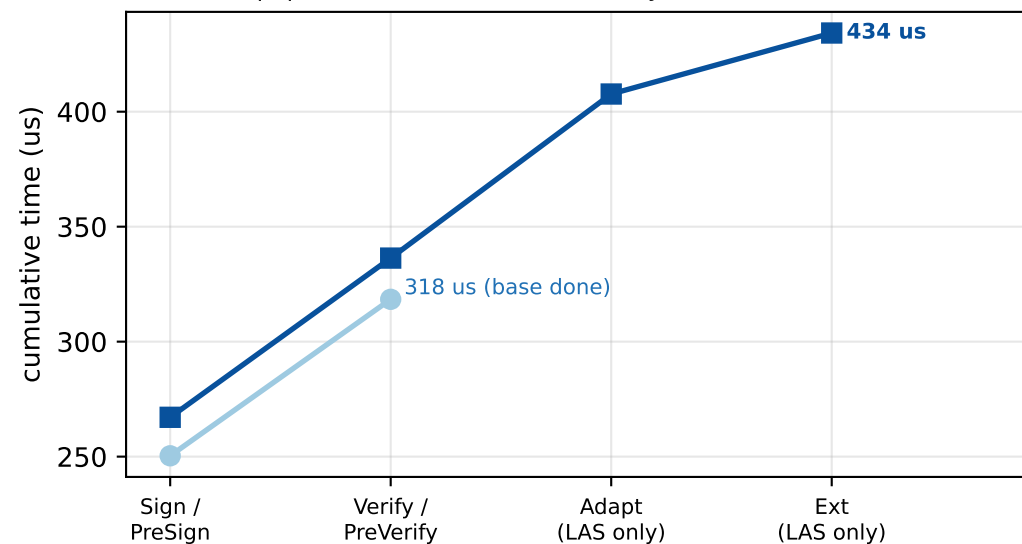
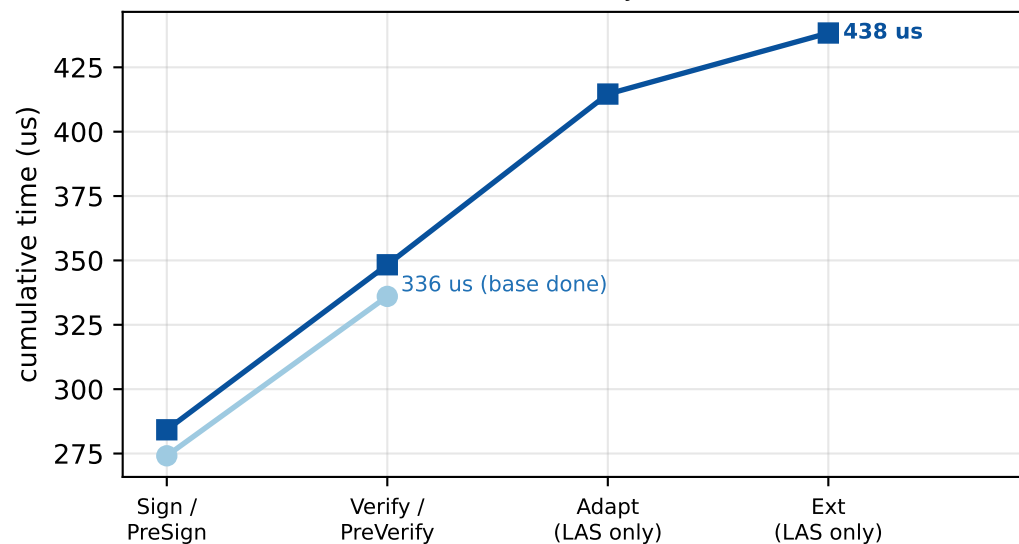


Protocol-step timeline: LAS adaptor cycle vs base signature (named steps; rejection-sampling Sign/PreSign dominates each path)

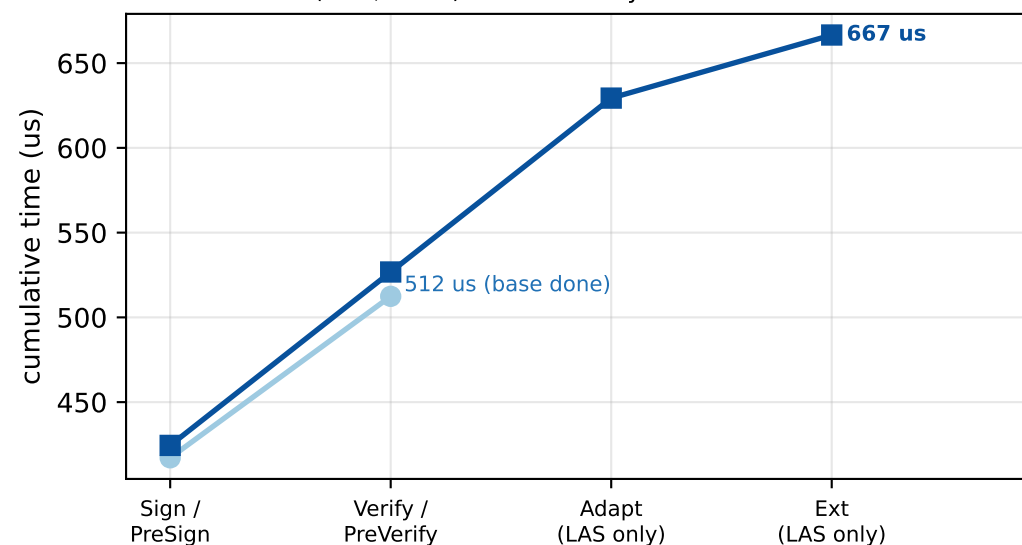
paper (n=4, ell=4) — full LAS cycle +36.4% vs base



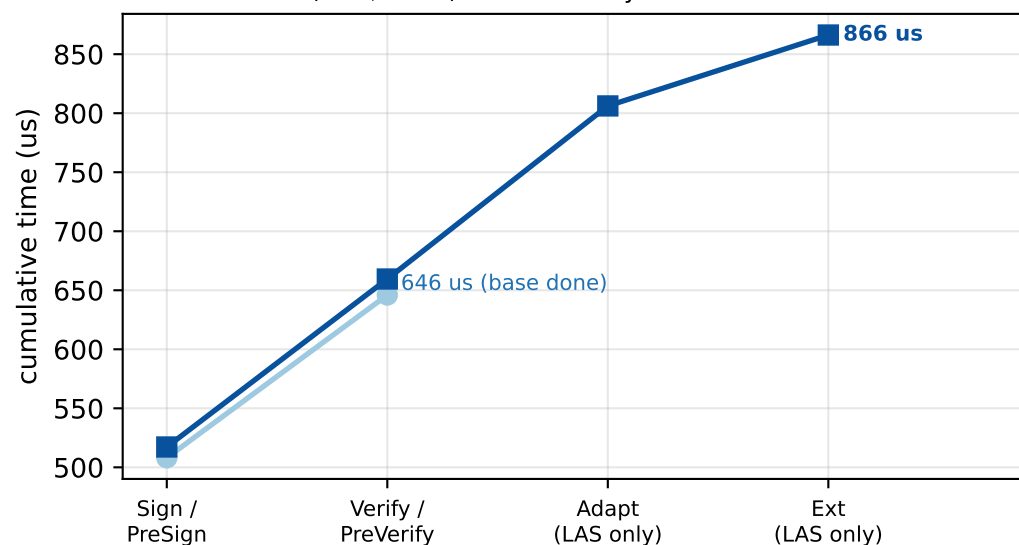
L2 (n=4, ell=4) — full LAS cycle +30.4% vs base



L3 (n=6, ell=5) — full LAS cycle +30.1% vs base



L5 (n=8, ell=7) — full LAS cycle +34.1% vs base



—●— Base path (Sign→Verify) —■— LAS path (PreSign→PreVerify→Adapt→Ext)